

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			O ₂ on products side (1) 2 in box (1) balancing mark only awarded if O ₂ correct		2		2	1	
	(b)	(i)		catalyst 1 is least effective because it has the lowest volume of gas collected after any given time / has the slowest reaction			1	1		1
		(ii)		all points plotted correctly (2) 4/5 points plotted correctly (1) tolerance $\pm \frac{1}{2}$ square curve through points from origin (1) ecf possible if plotting errors		3		3	3	
		(iii)		steeper curve (1) reaching final volume of 80 (1)		2		2	1	2
		(iv)		can replace bung / delivery tube before reaction starts (1) no loss of gas (1)			2	2		2
				Question 5 total	0	7	3	10	5	5